MAPLE ZHANG

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EDUCATION

University of Illinois Urbana-Champaign Aug. 2022 - Dec. 2024
Master of Computer Science GPA: 3.56/4.0
Research Area: Data Systems, Machine Learning, Autonomous Vehicles, Cloud Networks, Internet of Things

Shandong University Sept. 2017 - Jun. 2021
Bachelor of Engineering in Computer Science and Technology Overall GPA: 3.82/4.0
Relevant Coursework: Data Structures & Algorithms, Computer Networks, Database Systems, Operating Systems

TECHNICAL SKILLS

AI / Agent Engineering: Agent harness (Codex, Claude Code), multi-agent orchestration, MCP, Agent Skills, tool calling, RAG, temporal retrieval, Qdrant, SSE streaming, prompt engineering, Anthropic API, OpenAI SDK
Programming: Python, Go, TypeScript/JavaScript, SQL, C/C++/C#, Java, Shell, Solidity
Backend / Services: FastAPI, Gin, Node.js, Prisma, GORM, gRPC, REST, GraphQL, .Net, PostgreSQL, MySQL, Redis
Frontend / Product: React, Next.js, React Native, SwiftUI, Cesium, ECharts, Canvas, Tailwind CSS
Cloud / Delivery: Docker, Kubernetes, AWS (ECS, Lambda, RDS), GCP, Azure, Fly.io, Vercel, GitHub Actions, Playwright

PROFESSIONAL EXPERIENCE

Space AIC Mar. 2026 – Present
Software Engineer, Shanghai, China

- Developed **SAT Agent** for satellite Q&A and simulation support and a digital-twin copilot that turns natural-language queries into parameterized simulation runs over a live constellation model.
- Designed a space–ground **edge-AI architecture** for on-satellite large-model inference: model placement, bandwidth-constrained scheduling, and ground-side fallback.
- Built **SDT Education**: a Notebook–Python engine–Express API–Cesium workflow with per-student JupyterHub/Docker labs, closing the model–code–simulate–visualize–grade loop.
- Integrated the control plane with a **distributed RPC simulation backend** (topology, weighted paths, sharded simulators) and delivered hybrid-in-the-loop **microVM terminals** through a Terminal Gateway over HTTP/WebSocket/gRPC.
- Scaled the world's first 10,000+ satellite digital twin from 1,800 to **15,000+** satellites with interactive 3D playback (React 19, TypeScript, Cesium, Zustand); built Cell/Beam rendering for 1,584-satellite experiments.

Nethermind Aug. 2023 – Nov. 2023
Internship Program @Ethereum Foundation, Remote

- Migrated Voyager, a StarkNet blockchain explorer, from AWS Lambda and the StarkNet gateway to AWS ECS and RPC endpoints, rewriting 2 APIs and documenting the new architecture.
- Refactored the search module to route users to appropriate search paths by input type, reducing search latency by $\sim$1,000 ms.
- Improved AWS RDS retrieval efficiency by adding indices and rewriting PostgreSQL queries.
- Devised a **prompt-engineering** method to auto-detect and decode Cairo types into human-readable transaction call-data, improving accuracy over regex matching.

Microsoft Jan. 2022 – Aug. 2022
Software Engineer, Suzhou, China

- Designed and implemented a seamless PC-Phone pairing flow via QR/Accessibility code with 8 RESTful APIs and .Net Framework, increasing completion rate during Windows OOBE from 28.3% to 35.4%.
- Reduced server and database pressure via microservice architecture, asynchronous programming, caching, websockets, long polling, and API throttling, achieving $\sim$97% API success rate at 720 RPS.
- Built account-based cryptographic trust between PCs and phones by leveraging JWT encryption on certificate, device, and account information to ensure secure data transport and verification.
- Improved reliability and availability by adding over 2,000 lines of unit/functional/end-to-end/load tests and metric telemetry for anomaly detection, analytics, and troubleshooting.

Didi Global Jan. 2021 – Dec. 2021
Software Development Engineer, Beijing, China

- Refactored 6 APIs of a Point of Interest service serving **15 million users/day** from PHP to Go; CPU idle improved from 36% to 73% and TP99 latency dropped 51% at 2,000 QPS.
- Divided the business process into pluggable modules with an inner workflow engine and replaced 8 RESTful APIs with DiRPC plus Service Discovery for better SOA governance.
- Applied A/B testing to realize canary deployment, configuration synchronization, and strategy experiments; employed metrics components to trace logs generated during service context for troubleshooting.
- Managed online metric alarms (error ratio, latency) across 22 APIs and 16 downstream services while on-call.

Alibaba Cloud Jul. 2020 – Sept. 2020
Software Engineer Intern, Hangzhou, China

- Chaired a route-traceback project based on by-pass sniffing for SLA monitoring and network fault location.
- Constructed spoofed-source VIP packet sequences with increasing TTL fields to specific endpoints using Scapy.
- Employed traffic mirroring for packet capture and asynchronous I/O with call-back filtering.
- Delivered on Advanced Anti-DDoS clusters, elastically extendable via configuration files.

AI PRODUCTS (INDEPENDENT, SHIPPED)

Job Machine — Multi-Agent Job Hunter Sept. 2026
Open source at github.com/onemapl3/job-machine

- Built a **multi-agent** system that discovers, screens, applies to and follows up on jobs unattended: a coordinator derives actions from shared lead state and routes them to four role agents (Radar, Watch, Reacher, System); collected **1,780+ leads** from 44 companies in three weeks.
- Designed a **model-agnostic harness** where generation, agent executors and judgment are config-selected drivers with ordered chains and gateway fallback, so switching among headless **Claude Code**, **Codex** and OpenAI/Anthropic-compatible gateways is a JSON edit and tests run offline on fake drivers.
- Replaced prompt-and-parse matching with a small typed judgment model (TypeSafe Jev) that selects among code-extracted candidates and cites verbatim JD text, leaving large models to prose and browser work; re-judged **1,680 leads in 6.5 min** and cleared the discovery backlog.
- Packaged capabilities as **Agent Skills** and prompt-file workflows over a job queue, and added executor heartbeats that classify stalls (captcha, code, login, quota) and alert within 10 min, so scheduled cycles, a debounced chat executor and an hourly mail poll run without supervision.
- Turned every skip, pass and apply into time-decayed **preference weights** and learned prefilter rules, calibrated the judgment model against user feedback, and exposed the whole state through a local workbench, a Feishu Base and a Feishu bot with swipe screening and a chat grammar.

AmeBook — Agent-Driven Photo Book Generator Aug. 2026
Live at amebook.onemaple.cc, open source at github.com/onemapl3/AmeBook, shipped as a printed 240-page book

- Built a **local-first** photo-book generator that aggregates six source types, deduplicates a 90,000-image library, and references originals in place so no photo leaves the machine.
- Designed an **agent pipeline** to select, group, sequence, lay out, and caption an editable 240-page, 120-spread book; implemented 140 spread templates, 123 page layouts, 4 themes, and 4 trim sizes.
- Exported **300 dpi PDF with bleed** for direct printing and released the macOS client/open-source project (React, Canvas, Sharp, PDFKit, Electron) at github.com/onemapl3/AmeBook.

Type Machine — Multimodal Capture & Knowledge Pipeline Apr. 2026 – Present
Live at app.type-machine.com

- Built a mobile-first **Next.js App Router** PWA with Web Share Target ingestion, accepting screenshots, audio, video, and URLs from the system share sheet or desktop upload.
- Designed an asynchronous multimodal pipeline routing media through OCR, speech transcription, and multimodal processing into an editable-note schema without blocking uploads.
- Implemented ordered screenshot reconstruction with overlap deduplication, UI-noise filtering, language detection, and source-text preservation, eliminating the manual OCR copy-and-stitch workflow for long articles.
- Added AI post-processing for summaries, tags, and bilingual output, plus persisted task history, search/edit/review workflows, and Flomo API delivery so processed knowledge remains traceable and reusable.

Time Machine — Temporal-RAG Companion Agent Jan. 2026 – Mar. 2026
Deployed at timemachine-app.fly.dev, dual-agent SSE with date-bounded memory

- Engineered **temporal memory distillation and date-bounded RAG**, extracting dated anchors and injecting only memories preceding a chosen time point.
- Implemented **dual-agent** real-time SSE streaming with a SecondMe digital twin for dialogue and an Anthropic agent for structured analysis; shipped on Next.js 16, Prisma/SQLite, OAuth2, and Fly.io.
- Designed a Canvas/requestAnimationFrame **TimePortal state machine** from onboarding through relationship and memory selection to time travel and conversation.

OnePoem — Cross-Platform Composition Tool Apr. 2026
Live at onepoem.vercel.app, Web + iOS

- Created a mobile-first **React** poetry editor with reusable templates, typography, layout ratios, collections, and widgets; PNG export via html2canvas with an SVG fallback.
- Packaged it as a **SwiftUI/WKWebView** iOS app bridging JavaScript to native Photos saving and system sharing.